

MFU WIFI MAP

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Abstract— MFU WIFI MAP is a web service which lets users monitor access points devices in Mae Fah Luang university to track location and status of access points in university through a website. The main purpose of this project is to provide superiority for each user and network staff in the university. The Simple Network Management Protocol (SNMP) is used to communicate between access points and web services, which is used to show status and some data from access points to web services.

Keywords— Status Checking, Map, Wi-Fi, simple network management protocol, ISO Object Identifier

I. INTRODUCTION

Nowadays the internet is an important system which is used to connect people around the world. The internet can be used for many purposes. Some people use it for entertainment, some use it for education and some use it to keep in touch with their important person. To make it happen, the internet requires many kinds of network devices. Mae Fah Luang university also provides network devices and internet connection to give people inside university access to the internet.

Mae Fah Luang university contains many types of network devices such as network controller, router, switch layer 2, switch layer 3, servers, access point, etc. in this project we are focused on access point, to monitoring status, data and location of each access point, which provides advantages for staff to detect and fix failed devices in time.

II. RELATE WORK

A. Design and Implementation of Server Monitoring System Based on SNMP

The paper which introduces how to monitor servers through simple network management protocol (SNMP). To expand MIB resources by defining MIB objects to monitor the resources of servers, and use multi-threading technology to collect data and process, which can improve the collection efficiency. The experimental results prove a successful way of integrated monitoring and control for servers based on SNMP.

B. Chiang Mai University's Jumbo Map

The web service of Chiang Mai University which is used in purpose to locate and display status results of access points around the university, the utilization of Google map is used to display areas around university and locate 5857 amount of access points in the form of markers. Besides, each marker

displayed in multiple colors, each color used to appraise the status of access points, and markers also reveal details of access points such as name and associated users' graph.[1]

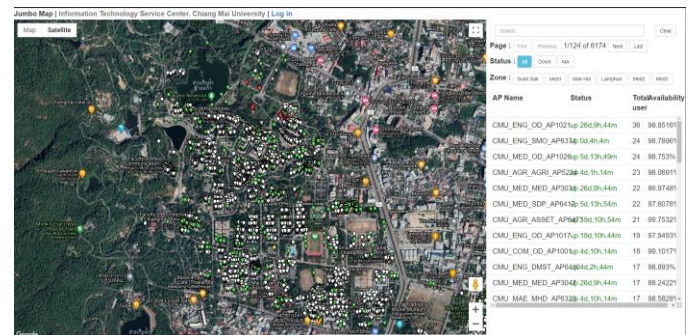


Figure 1 Jumbo Map

C. Land use information release system based on Google Maps API and XML

Web map service is a new map service mode actualized by the use of internet technology, computer technology and GIS technology. This article applies Web map service to the release of land use information, and tries to use Google Maps API and XML to build a network release system of regional land use information. The system integrates Google Maps data and user data effectively, carries out the display and management of land use information, and has a powerful public participation function. The study shows that Web map services based on Google Maps API and XML can play an important role in the release of land use information, which is also important in the information-based construction of land resources.

D. Access Point

The functions of an access point that mainly serves to distribute the Wi-Fi signal, which is popular to be placed at various points the radius of the Wi-Fi signal is covered so that the Wi-Fi compatible devices connect to the same LAN, so it is often seen that most APs only have one LAN port in order to connect. With the Network Switch or may be connected to the router at all to access the Internet of the service provider, but some brands of APs such as LINK have 2 ports with speeds of 10/100/1000 Mbps and 10/100 Mbps is provided as an additional option for customers to use. It also supports PoE.

An Access Point is a Layer 2 device and therefore works on Layers 1 and 2. Now, a Wireless Gateway (WG) usually comes with Limited Layer 3 Capabilities as it "handles" IP addresses, Protocol Filters, etc. Some may work with RIP (v1 and v2). A WG is an All-in-one product with an Access Point for Layer 2 Wireless Access. Moreover, Access Points are half-duplex devices with switching-like capabilities.[2]



Figure 2 Access Point

E. The Simple Network Management Protocol (SNMP)

A standard protocol for collecting and organizing information about devices in an IP Network when they are changed. SNMP-enabled devices, for example, modems, routers, switches, servers, workstations, printers, and other IT devices.

SNMP is widely used for network management or monitoring systems. SNMP stores data in the variable format and is managed by the management information base (MIB), which is a storage or database for managing or changing values on devices in the network at the moment. There are 3 versions of SNMP released. SNMPv1 was the first protocol and was developed to SNMPv2c and SNMPv3 respectively to increase the capability. Flexibility in use and security. SNMP consists of 3 main ones:[3]

- The devices in the network.
- Agent - This software must be installed on the device to detect and report anomalies.
- Network management station (NMS) - SNMP server-side software, also known as manager, receives and collects information from agents.

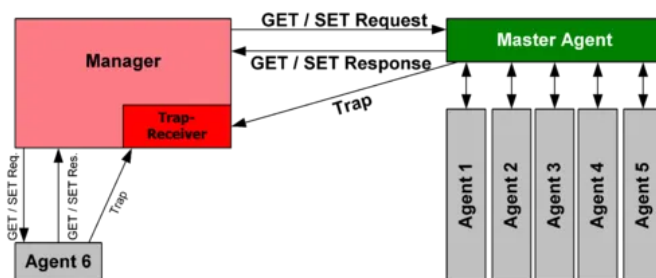


Figure 3 Processing of SNMP

F. Google map

Google Maps covers all applications and is the largest web mapping service. Users with just an Internet connection will be able to use the real-time Google Maps service. And also used to plan travel. Used to check the distance of the road. And used to check the space of the desired area. Check width and length of different areas. To make the location search. The names of streets and intersections are fast and convenient. Especially restaurants.

Google Maps users can view and see maps. Around the world easily and quickly. Can make or enlarge the map around the world to be smaller. In addition, Google Map helps plan travel routes to various areas. Or you can drop a pin to map directions to your favorite places as well.[4]



Figure 4 Google Maps

G. Cisco Wireless LAN Controller (WLC)

Wireless Controller will help increase the distance of network connection. Including network cabling in hard-to-reach areas or unable to install easily, the Access Point itself also provides a solution about the network installation. With a small budget. And Cisco wireless solutions help secure wireless communications. Can be handled easily. It is also stable to operate at any speed and distance.

There are two types of Cisco wireless solutions:

- An Access Point device that is directly connected to a wired network.
- 2-part connection required to depend on controller device with access point directly connected to controller or device with centralized control. And this device will connect to the wired network again.



Figure 5 Cisco Wireless LAN Controller

For convenience and ease of management, we need a WLC to control all access points. We can manage all access points through WLC. Cisco access points that can be used with Cisco WLC are called Lightweight Access Point (LAP). We will be able to control everything of an access point through WLC, be it Channel, Roaming, Transmission Power, etc., as well as various configurations of the access point. To register or join with WLC.[3]

III. MEDTHODOLOGY

The working of this project is separated into two parts. The first part is a web site which is used to display map, markers and access points status. The second part is made up of network controller and python programming which is used to send SNMP requests to the network controller to retrieve connected access points data. The project uses MySQL database to store retrieved data for web service.

A. Preprocessing

The project starts with collecting access points OID number to database (OID number of access points is used for identification access point). The OID number of access points will be used to connect with the OID number of SNMP walk protocol to retrieve specific data from each access point.

1.3.6.1.4.1.14179.2.2.1.1.3	bsnAPName
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Figure 6 OID number of access point's name

B. Retrieve Access point's data

The project sends SNMP walk requests to access point devices to collect data. To send large amounts of SNMP walk requests the project uses python programming to query commands through the Linux server's terminal. The network controller will reply data of connected access points to the python program for analysis and update into the database.

OID	name
0.41.194.219.122.32	S80167-MFU-F3F04-AP651
0.41.194.219.123.0	S80167-MFU-ST1F02-AP956

Figure 7 Access point's OID number in database.

```
"snmpwalk -c 192.168.1.1 -v1 iso.3.6.1.4.1.14179.2.2.1.1.3."
```

Figure 8 SNMP walk command with OID number for access point's name

C. Store analyzed data

The project keeps a large amount of data in the MySQL database by using python MySQL connector. The database will contain access point's OID number, name, latitude, longitude, status, and associated users, which have been updated by python.

OID	name	lat	lng	status	user
0.41.194.219.122.32	S80167-MFU-F3F04-AP651	20.049788	99.893883	up	2
0.41.194.219.123.0	S80167-MFU-ST1F02-AP956	20.049885	99.892609	up	8
0.41.194.219.129.192	S80167-MFU-LD4F02-AP924	20.058310	99.898369	up	1
0.41.194.219.133.128	S80167-MFU-F3F03-AP659	20.050026	99.893753	up	2
0.41.194.219.136.32	S80167-MFU-F3F01-AP638	20.049686	99.893631	up	0
0.41.194.224.210.64	S80167-MFU-LD5F03-AP847	20.057629	99.897751	up	4
0.47.92.102.146.96	S80167-MFU-LD5F03-AP849	20.057806	99.897919	up	1
0.47.92.102.211.96	AP002F-SC66.2428	20.046268	99.893562	up	0

Figure 9 Data inside the database

D. Insert Latitude and Longitude.

To insert latitude and longitude, the project uses Google Maps API function get lat/lng. This function allows the user to know latitude and longitude when clicking that position on a map. Then, update the database.

Access Points DATA

Go to WiFIMAP Page Logout

Search for names...

OID Number	Access Point Name	Latitude	Longitude	Action
0.47.92.102.211.96	AP002F-SC66.2428	20.046268	99.893562	Update Delete
0.182.112.101.191.96	RE203	20.046356	99.893402	Update Delete
88.172.120.235.65.176	S80167-AVF01-AP01	20.046980	99.893166	Update Delete
88.172.120.199.156.128	S80167-AVF01-AP02	20.046980	99.893250	Update Delete
204.70.214.8.14.224	S80167-AVF01-AP03	20.046974	99.893356	Update Delete
204.70.214.8.23.144	S80167-AVF01-AP04	20.046974	99.893425	Update Delete
88.172.120.235.134.48	S80167-AVF02-AP01	20.046925	99.893158	Update Delete

Figure 10 Access point data table

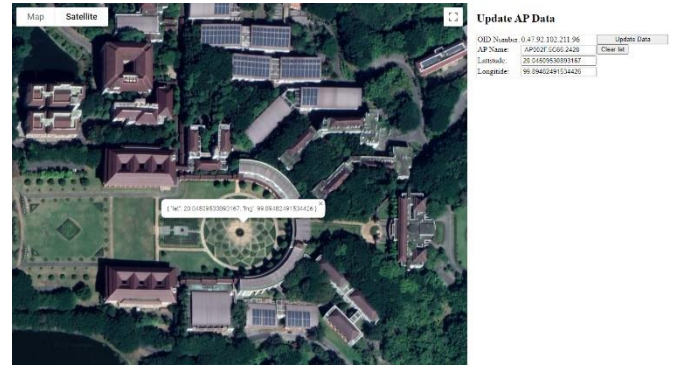


Figure 11 Get Lat/Lng function from Google Maps API

E. Website Design

Google Maps API is used to display all devices location in marker form, each marker has indicated color which shows device status. The red marker indicates problems on the device, white markers indicate operated devices and no associated user, the green marker indicates associated user between 1 to 20, the yellow marker indicates associated user is more than 20, the orange marker indicates associated users is more than 40. The SQL database is used to obtain data on devices and display on the left side of the website.

The website also has administrative and staff accounts in purpose to manage and update access points. The administrator interface can be found in figure11.

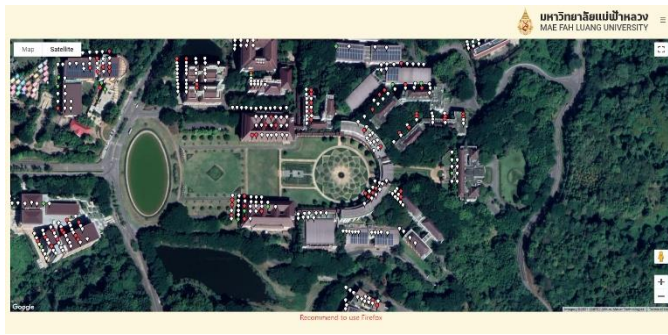


Figure 12 Web design

IV. IMPLEMENTATION

The project requires a Linux OS server, in order to handle processes of the project which are python scripts and website scripts. To present data from the access point accurately the project will be updated every 5 minutes and collect data of each update for analysis in the future. The python programming is scheduled to work properly even if the server will be rebooted.

Website administrators have authority to update location or remove access points from the project, while normal users are able to see only the index page. The access points location must be manually assigned to the database through the access points configuration website.

V. SUGGESTION

This project can be developed to create a wider variety of applications. In addition to viewing the status and number of

people access, or to improve various aspects of the Access Point's installation plans in the future, for example: Developed in the safety of communication devices of students by programming, access to the location of communication devices of students or staff. Provided that the device must be connected to one of the university's Access Points. Accessing this location will enable us to find communication devices for students and staff if they are lost by checking the latest connection of communication devices of students or staff. However, this project is a starting point for students who want to develop a website or program to meet the needs of solving various problems. Which we have presented here may not solve many problems that may arise in the future. But we hope that this project will be a way of making projects for future generations.

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